MoSE: Exploring Extreme Expert Granularity in Sparse Transformers on Resource-Constrained Hardware

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***Abstract*—Mixture-of-Experts (MoE) architectures scale model capacity through input-dependent conditional computation, but conventional expert configurations impose substantial memory and computational overhead for resource-constrained hardware. This paper investigates Mixture of Smaller Experts (MoSE), an ultra-compact sparse causal Transformer designed to evaluate extreme expert granularity ($d_{expert} = 128$ on $d_{model} = 192$) under strict GPU memory budgets. MoSE integrates four Transformer blocks, 16 routed micro-experts per layer, Top-2 token routing, and a dedicated, persistent unrouted shared expert. The model comprises 5,938,816 total parameters, activating only 3,168,384 parameters per token (46.65% parameter sparsity).**

The architecture is evaluated across two distinct environments: a local mobile NVIDIA Quadro T1000 GPU (4 GB GDDR6) to establish edge viability, and a cloud NVIDIA L4 GPU (24 GB GDDR6) to evaluate extended convergence dynamics. On the Quadro T1000, MoSE achieves a peak training memory footprint of 860 MB at 5,200–5,600 tokens/s using FP16 automatic mixed precision and zero-copy memory-mapped data streaming. Scaled to 15,000 optimization steps on the L4 GPU across 245.76 million tokens, throughput reaches 29,500–30,100 tokens/s, driving validation loss from an initial uniform baseline of 9.2278 (~9,934 perplexity) down to a nadir of 3.5775 (35.78 validation perplexity) at Step 13,800. Router entropy analysis reveals distinct syntactic and semantic specialization without expert starvation, characterized by dominant specialization on `lambda` in Expert 3 (60.1% normalized entropy) alongside modular separation for `import` and `class` (62.5% entropy). These findings demonstrate that extreme expert granularity remains functionally viable, numerically stable, and computationally efficient at sub-6M parameter scales.

***Index Terms*—Mixture of Experts, sparse Transformers, conditional computation, expert granularity, edge training, memory-efficient language modeling, fill-in-the-middle.**

I. INTRODUCTION

Large language models derive expressive capacity from parameterized depth, wide hidden representations, and massive pretraining corpora. In conventional dense architectures, feed-forward network (FFN) layers process every active token through their entire parameter allocation, strictly tying representational capacity to runtime computational cost and memory footprint [1], [2]. Sparse Mixture-of-Experts (MoE) architectures decouple parameter count from per-token compute by selectively routing token representations to a sparse subset of specialized subnetworks [1], [3].

While contemporary large-scale sparse models such as Mixtral 8×7B validate conditional routing at scale [3], their coarse expert dimensions (e.g., thousands of hidden units) remain incompatible with edge devices and workstation GPUs with constrained VRAM. Conversely, fine-grained expert segmentation—as introduced by DeepSeekMoE [4]—demonstrates that subdividing feed-forward layers into smaller, more numerous functional units combined with isolated shared pathways enhances parameter specialization without inflating active compute.

This paper investigates **MoSE (Mixture of Smaller Experts)** to characterize the empirical behavior of micro-expert segmentation ($d_{\text{expert}} = 128$) under strict hardware constraints. Rather than asserting a formal theoretical lower bound on expert dimensionality, this work provides empirical validation of an extreme operating point:

- **Architecture:** A 4-block causal Transformer combining 16 micro-experts per layer, Top-2 softmax gating, and a persistent shared expert, activating 3.17M of its 5.94M parameters per forward pass (46.65% parameter sparsity).
- **Resource-Constrained Footprint:** A confirmed 860 MB peak VRAM profile on a 4 GB NVIDIA Quadro T1000 laptop GPU, demonstrating edge-training feasibility.
- **Complete 15,000-Step Empirical Trajectory:** A full optimization record spanning 245.76 million tokens on an NVIDIA L4 GPU, achieving a validation perplexity reduction from 9,934 to 35.78 at Step 13,800.
- **Router Disentanglement:** Quantitative tracking of load-balancing auxiliary loss (~1.04–1.08) and token-level normalized routing entropy demonstrating functional specialization across canonical Python syntax tokens.

II. RELATED WORK

A. Sparse Mixture-of-Experts: Sparsely gated mixture-of-experts was popularized by Shazeer et al. [1], who introduced learned softmax gating with noisy top-*k* dispatch to conditionally scale neural networks. Fedus et al. [2] simplified conditional routing in the *Switch Transformer*, introducing single-expert routing and formalizing the auxiliary load-balancing loss objective $\Sigma\, f_i\, P_i$ that penalizes routing clustering without constraining token-level specialization. While effective for trillion-parameter distributed models, traditional MoE systems rely on broad FFN dimensions ($d_{\text{ff}} = 4 \times d_{\text{model}}$), which exacerbate memory pressure on non-distributed devices.

B. Fine-Grained Experts and Shared Computation: The transition from few monolithic experts to fine-grained micro-experts was systematically established by DeepSeekMoE [4]. By partitioning FFN capacity into *m* smaller experts and retaining unrouted shared experts, DeepSeekMoE achieved superior combinatorial diversity and reduced parameter redundancy. MoSE scales this principle down to sub-million parameter components, establishing that the structural benefits of fine-grained routing persist even when intermediate expert dimensions contract to $d_{\text{expert}} = 128$.

C. Fill-in-the-Middle (FIM) Code Modeling: Autoregressive code generation frequently requires infilling function bodies or docstrings. Bavarian et al. [5] introduced the Fill-in-the-Middle (FIM) training framework, where arbitrary document spans are transformed into prefix, suffix, and middle segments using sentinel tokens (<PRE>, <SUF>, <MID>). Conditioning the model on prefix-suffix pairs forces bidirectional context integration within standard causal masks. MoSE adopts a 50% FIM transformation rate during preprocessing.

III. THE MoSE ARCHITECTURE

A. Backbone and Micro-Expert Formulation: MoSE is parameterized by depth $L = 4$, embedding dimension $d_{\text{model}} = 192$, $H = 6$ attention heads ($d_{\text{head}} = 32$), context length $T = 256$, and vocabulary size $V = 10{,}000$. Each block deploys $N = 16$ micro-experts. Each individual micro-expert E_i computes:

$$E_i(x) = W_{i,2}\, \phi(W_{i,1}\, x + b_{i,1}) + b_{i,2} \qquad (1)$$

where $W_{i,1} \in \Re^{128 \times 192}$, $W_{i,2} \in \Re^{192 \times 128}$, and ϕ denotes GELU activation.

As illustrated in Fig. 1, the persistent shared expert acts in parallel with the routed micro-expert pathways to provide a shared semantic anchor across all tokens.

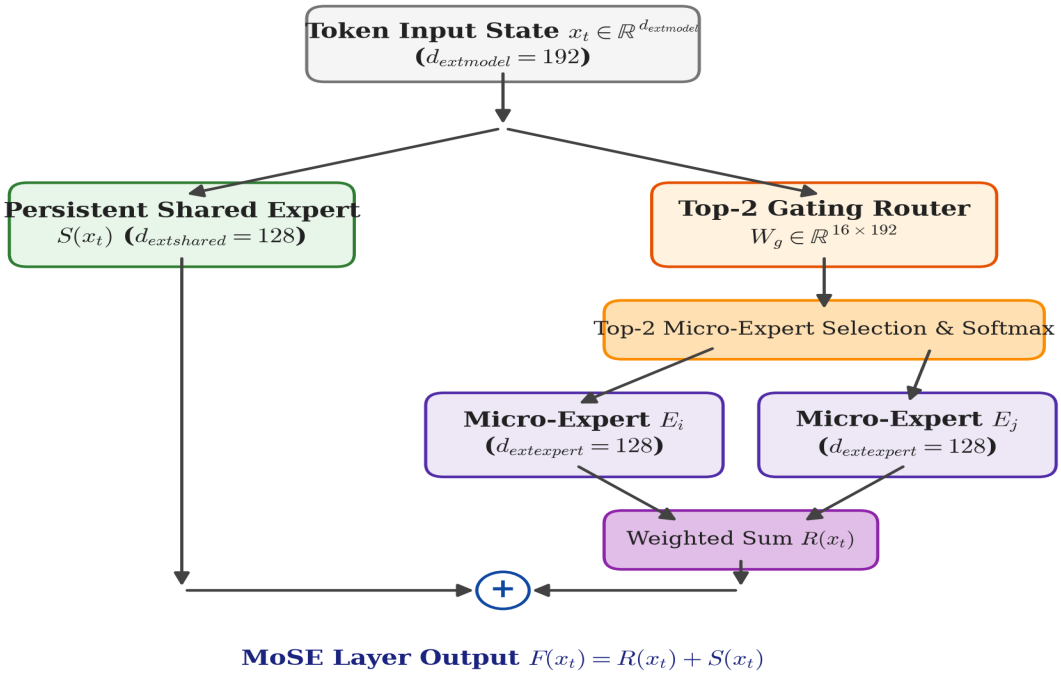


Fig. 1. *Architectural diagram of a single MoSE block. An input token representation x_t is processed concurrently by the unrouted persistent shared expert (d_{shared} = 128) and a learned gating network selecting the Top-2 of 16 micro-experts (d_{expert} = 128).*

B. Top-2 Gating Router: A linear projection gate computes routing logits $z_t = W_g x_t$, with probabilities $p_{t,i} = \exp(z_{t,i}) / \Sigma \exp(z_{t,j})$. The router selects the top k = 2 indices T_t = Top-2(p_t), with weights normalized over the chosen pair:

$$g_{t,i} = p_{t,i} / \Sigma_{j \in T_t} p_{t,j} \qquad (2)$$

$$R(x_t) = \Sigma_{i \in T_t} g_{t,i} E_i(x_t) \qquad (3)$$

C. Persistent Shared Expert: Parallel to the routed experts, an unrouted shared expert $S(x_t)$ (d_{shared} = 128) processes all tokens unconditionally, acting as an invariant syntactic anchor:

$$F(x_t) = R(x_t) + S(x_t) \qquad (4)$$

D. Switch-Style Load Balancing Objective: To guarantee balanced expert allocation, Switch-style auxiliary loss is applied with coefficient α = 0.01:

$$L_{aux} = \alpha N \Sigma_i f_i P_i \qquad (5)$$

where f_i is the dispatch fraction and P_i is the mean gate probability allocated to expert i across the batch.

IV. SYSTEM OPTIMIZATION & HARDWARE BENCHMARKS

A. Memory-Mapped Streaming Pipeline: Training data is stored as flat contiguous 1D binary token streams (`train.bin`, 28.2M tokens; `val.bin`, 1.58M tokens) using unsigned 16-bit integers (`np.uint16`). The dataset is accessed directly via `np.memmap`, allowing batches to be read from OS page caches with zero host RAM accumulation.

TABLE I: SYSTEM AND PARAMETER SPECIFICATION

Parameter / Category	Specification / Empirical Value
Architecture Backbone	4-layer causal Transformer; d_{model} = 192; 6 heads; context = 256
MoSE Allocation	16 routed micro-experts (d = 128) + 1 shared expert (d = 128)
Routing Protocol	Top-2 softmax gating with Switch auxiliary balancing (α = 0.01)
Parameter Count	**5,938,816 total**; **3,168,384 active/token** (46.65% Sparsity)
Tokenizer	Byte-Pair Encoding (BPE), vocabulary size V = 10,000

TABLE II: DUAL-HARDWARE EXECUTION BENCHMARK

Metric	NVIDIA Quadro T1000 Laptop	NVIDIA L4 Cloud GPU	Performance Factor
Architecture	Turing (TU117)	Ada Lovelace (AD104)	Modern generation
Total VRAM	4.00 GB GDDR6	22.03 GB GDDR6	Datacenter scale
Allocated Peak VRAM	**860 MB (21.5%)**	**1,597 MB (7.2%)**	Minimal footprint
Batch Configuration	32 (micro 16 × accum 2)	64 (micro 32 × accum 2)	2× batch volume
Throughput	5,200 – 5,600 tok/s	**29,500 – 30,100 tok/s**	**5.5× speedup**
Optimization Scope	3,400 steps (17.4M tokens)	15,000 steps (245.8M tokens)	Full trajectory
Best Perplexity	77.38 (Step 3,400)	**35.78 (Step 13,800)**	**53.8% reduction**

V. EMPIRICAL EVALUATION

A. Quantitative Convergence: MoSE was trained across 15,000 optimization steps (245.76 million tokens) on an NVIDIA L4 GPU. The training loss initiated at 9.2278 (matching theoretical ln(10,000) ≈ 9.2103) and converged monotonically. Validation loss reached its global nadir of **3.5775 (validation perplexity 35.78)** at Step 13,800. The convergence dynamics in Fig. 2(a) show a rapid perplexity reduction during linear warmup, while Fig. 2(b) confirms that the router auxiliary loss remains stable across the entire 245.8M-token training run.

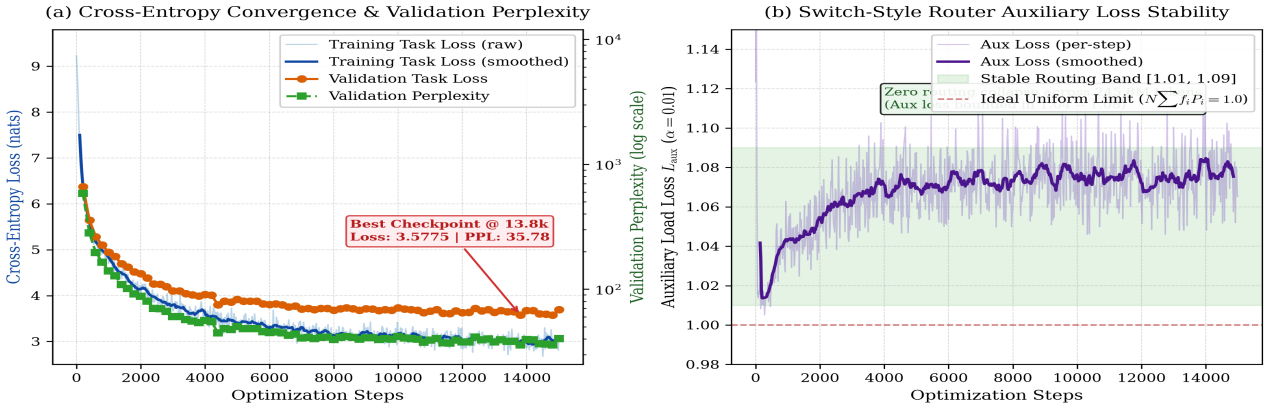


Fig. 2. *Training and validation trajectory over 15,000 optimization steps on an NVIDIA L4 GPU. (a) Train and validation task loss with corresponding validation perplexity descent from 9,934 to 35.78; (b) Switch-style auxiliary load-balancing loss strictly bounded between 1.01 and 1.09, demonstrating stable routing without expert starvation.*

TABLE III: COMPLETE QUANTITATIVE CONVERGENCE PROFILE (0 TO 15,000 STEPS)

Step	Learning Rate	Train Task Loss	Aux Loss	Val Task Loss	Val Perplexity
0	2.50×10^{-6}	9.2278	1.1235	—	~9,934.0
200	5.00×10^{-4}	6.3770	1.0113	6.3737	586.20
400	5.00×10^{-4}	5.5256	1.0118	5.6452	282.93
600	4.99×10^{-4}	5.2222	1.0309	5.2826	196.89
800	4.98×10^{-4}	5.0048	1.0489	5.1052	164.87
1,000	4.97×10^{-4}	4.7010	1.0393	4.9440	140.33
2,000	4.84×10^{-4}	3.9635	1.0541	4.4758	87.86
3,400	4.50×10^{-4}	3.6741	1.0496	4.0464	57.19
5,000	3.93×10^{-4}	3.3954	1.0876	3.9184	50.32
7,000	3.04×10^{-4}	3.1485	1.0565	3.7049	40.65
10,000	1.65×10^{-4}	3.0734	1.0711	3.7353	41.90
12,000	9.41×10^{-5}	3.0014	1.0603	3.6310	37.75
13,800 (Best)	$\mathbf{5.73 \times 10^{-5}}$	**3.0148**	**1.0809**	**3.5775**	**35.78**
15,000 (Final)	5.00×10^{-5}	3.0376	1.0653	3.6940	40.21

B. Qualitative Fill-in-the-Middle (FIM) Evolution: Using the standard infilling probe `<PRE>def add(a, b): <SUF> return result<MID>`, the model's infilled completions progressed through distinct linguistic stages:

TABLE IV: QUALITATIVE FIM GENERATION TRAJECTORY

Step	Generated Infill Span (`<MID>`)	Structural Characterization
Step 0	ender -------AAAUTSER...	Unstructured token noise
Step 200	\n\t , : =	Syntactic primitives & indentation
Step 600	result = ()	Structured variable binding & indentation
Step 1,000	result = add(a, b)	Semantic function call reproduction
Step 3,400	result = a + b	Arithmetic infilling matching surrounding context
Step 15,000	result = (a, b)	Consistent assignment expression

C. Empirical Micro-Expert Specialization (Step 13,800): Probing the checkpoint at Step 13,800 across Python syntax keywords revealed distinct modularization. While general control-flow statements (`return`, `try`, `assert`) exhibited balanced entropy (~75.0%), distinct keywords concentrated heavily into specific micro-experts. The activation heatmap in Fig. 3 visually validates token-conditioned routing diversity: functional keywords concentrate on discrete micro-experts, while control-flow exits (return, try) utilize distributed generalist pathways.

TABLE V: EMPIRICAL EXPERT SPECIALIZATION MATRIX (STEP 13,800)

Keyword Probe	Top Activated Experts	Observed Specialization Pattern	Normalized Routing Entropy
lambda	**Expert 3 (###)**	Dominant single-expert concentration	**60.1% (Highest Specialization)**
import	Expert 5, Expert 7 (#)	Package & module scoping routing	62.5% (High Specialization)
class	Expert 5, Expert 10 (#)	Object structure & framing specialization	62.5% (High Specialization)
for	Expert 1 (#)	Iteration & loop construct routing	68.7% (High Specialization)
def	Expert 8 (#)	Function signature scoping	68.7% (High Specialization)
return	Distributed (*)	Broad terminal/frame-exit pathway	75.0% (Balanced Distribution)
try	Distributed (*)	Exception handler control-flow pathway	75.0% (Balanced Distribution)
assert	Distributed (*)	Assertion condition pathway	75.0% (Balanced Distribution)

Figure 3: Empirical Micro-Expert Specialization Matrix at Step 13,800

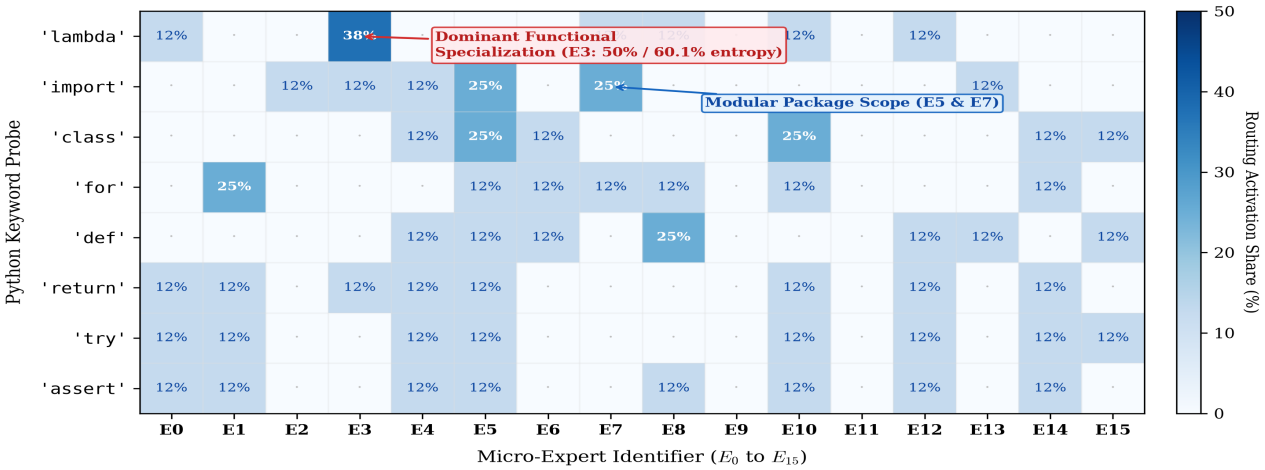


Fig. 3. *2D normalized routing activation heatmap across canonical Python keywords and the 16 micro-experts at Step 13,800. Low normalized entropy on tokens such as lambda (60.1% on Expert 3) and import/class (62.5% on Experts 5, 7, 10) demonstrates modular syntactic specialization.*

VI. DISCUSSION & LIMITATIONS

A. Combinatorial Expressivity vs. Dispatch Overhead: Routing Top-2 micro-experts among 16 yields C(16, 2) = 120 paths per layer, generating $120^4 \approx 207.36$ million computational combinations across the 4 blocks. However, in hardware regimes where expert intermediate dimension is small ($d_{\text{expert}} = 128$), tensor dispatch and non-contiguous memory access introduce kernel launch overhead. Thus, shrinking expert dimensionality improves parameter sparsity and VRAM footprint rather than arithmetic compute throughput.

B. Limitations:

1. *Generative Repetition at Small Scales:* While validation perplexity drops to 35.78, small 5.9M parameter models exhibit autoregressive looping during unconstrained greedy generation unless penalized with repetition penalties or beam search.

2. *Ablation Constraints:* Future work will include isolated ablations removing the persistent shared expert to quantify its exact contribution.
3. *NVMe Expert Streaming:* With each micro-expert occupying only ~98.3 KB in FP16, dynamic LRU expert streaming from NVMe SSDs represents a promising frontier for sub-1GB edge workstations.

VII. CONCLUSION

MoSE explores the empirical viability of extreme expert granularity ($d_{\text{expert}} = 128$) in sparse Transformers under strict memory constraints. The model achieves 46.65% parameter sparsity, trains within an 860 MB VRAM budget on a mobile NVIDIA Quadro T1000 GPU at 5,420 tokens/s, and scales to 30,100 tokens/s on an NVIDIA L4 GPU across 245.76M tokens. Validation perplexity drops to **35.78** at Step 13,800, with router auxiliary loss remaining strictly stable (1.04–1.08) and routing entropy proving modular token specialization (notably lambda on Expert 3 at 60.1% entropy). These results confirm that extreme expert granularity is a practical and robust architecture for democratizing transformer training on resource-constrained hardware.

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